

Xinhui CHEN

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EDUCATION

South China University of Technology

Guangzhou, CN

M.Eng. of Safety Engineering

Sep. 2022 – Present

- **Average Score:** 87 / 100
- **Core Courses:** *Case Studies on Production Accidents (91), Applied Statistics Theory and Method (93), Safety Production Technology (90), Fire Science (85)*

Northeastern University

Shenyang, CN

B.Eng. of Safety Engineering

Sep. 2018 – Jun. 2022

- **GPA:** 3.8 / 5.0 **Ranking:** 6 / 56 (Top 10.7%)
- **Core Courses:** *Mechanical Safety (97), Safety Measurement and Instrument (96), Architecture Safety (94), Emergency Technique (90), Safety Ergonomics (90)*
- **Course design:** *Fire and Explosion Prevention (A), Electric Safety (A), Accident Investigation (A)*

RESEARCH INTERESTS

- CO₂ Sequestration
- Unconventional Oil and Gas Extraction
- Computational Fluid Dynamics (CFD)
- Artificial Intelligence and Machine Learning
- Simulation, Modelling and Numerical Methods

PUBLICATIONS

- [1] Liu, Z. (Advisor), **Chen, X.***, Tian, F.*, Zhu, W., Hu, Z., Su, W., & Wang, Z. (2023). Dynamic effects of rough elements in fractures on coal permeability during different stages of methane extraction. *Gas Science and Engineering*, 119, 205118. DOI: 10.1016/j.jgsce.2023.205118.
- [2] **Chen, X.**, Huang, S., Guan, T., Fu, H., & Pan, Z. (Accepted). An interpretable predictive model for liquid holdup of hilly-terrain oil-gas pipelines based on machine learning and Shapley additive explanations. *ACS Omega*
- [3] **Chen, X.**, Huang, S., Guan, T., Fu, H., & Pan, Z. (Accepted). Prediction model of liquid holdup of multiphase pipelines based on PSO-BP algorithm. *Chemical Engineering (China)*.
- [4] **Chen, X.***, Tang, Z., & Mo, Y. (2023). Design, Processing and Performance Testing of Extra-Large-Flow Centrifugal Model Pumps. *Journal of Physics: Conference Series*, 2610(1), 012022. DOI: 10.1088/1742-6596/2610/1/012022.
- [5] Yang, G., **Chen, X.***, Hu, Y., Huang, S., & Xu, H. (2024). Analysis of vibration characteristics of cryogenic insulated cylinders based on ANSYS. *Journal of Physics: Conference Series*, 2860(1), 012016. DOI: 10.1088/1742-6596/2860/1/012016.
- [6] Xie, H., Ou, C., **Chen, X.**, & Yang, G. (2023). Numerical analysis of the temperature rise at fast filling of hydrogen storage cylinder based on fluid-thermal-solid coupling. *Journal of Physics: Conference Series*, 2610(1), 012025. DOI: 10.1088/1742-6596/2610/1/012025.

PATENTS

- [1] Huang, S. (Advisor), **Chen, X.**, Guan, T., & Fu, H. (2024). Prediction method for liquid holdup of hilly-terrain oil-gas pipelines based on SSA-BP neural network. (Chinese Patent No. 202410626009.7)
- [2] Huang, S. (Advisor), **Chen, X.**, Hu, Y., & Yang, G. (2024). A finite element analysis method for vibration characteristics of vertical low temperature insulated cylinder system. (Chinese Patent No. 202411477988.0)

RESEARCH PROJECTS

AI-based predictive model for liquid holdup in hilly-terrain multiphase flow pipelines

Principal Investigator | Advisor: Prof. Si Huang

Jun. 2023 – Present

- Constructed 18,900 sets of hilly-terrain oil-gas pipelines considering angles and elevations
- Simulated the oil-gas flow under 81 working conditions using the multiphase flow simulation software OLGA
- Developed an AI-based prediction model, leveraging a dataset comprising 648,000 simulation results
- Established an interpretable model incorporating SHapley Additive exPlanations to unravel the global contributions and interactions of factors in the AI black box

Influence of fracture morphology on gas migration patterns in coal

Principal Investigator | Advisor: Prof. Zhengdong Liu

Mar. 2021 – Sep. 2023

- Derived a new numerical model of rough-walled fractures that innovatively simplifies the fracture surface
- Proposed a concept of effective fracture aperture that varies nonlinearly with the CH₄ extraction process
- Constructed a CFD model of methane extraction using COMSOL Multiphysics, in which the governing equations for gas diffusion and seepage influenced by the new numerical model were imported
- Analyzed the dynamic effect of roughness, stagger degree and shape of rough elements on CH₄ migration

Simulation and testing of vibration characteristics of cryogenic insulated cylinders

Principal Investigator | Advisor: Prof. Si Huang

May 2024 – Present

- Formulated a modal analysis model for the cryogenic insulated cylinder utilizing ANSYS
- Delved into the natural frequency trends of the cylinder across varying filling percentages

Fluid-thermal-solid coupling of carbon fiber full-wrapped hydrogen cylinders during fast inflating

Co-investigator | Advisor: Prof. Si Huang

Jan. 2023 – Jun. 2023

- Established a fluid-thermal-solid coupled CFD model for fast inflating of carbon fiber full-wrapped hydrogen cylinders based on ANSYS
- Verified that the maximum stress point on the cylinder corresponds precisely to the vulnerability pinpointed by acoustic emission signal mapping

Hydraulic model design and test of large capacity vertical centrifugal pump

Co-investigator | Advisor: Prof. Si Huang

Sep. 2022 – Jan. 2023

- Designed a model pump based on a prototype pump according to similarity principles
- Constructed physical models of centrifugal pumps utilizing CFTurbo
- Established the CFD model of centrifugal pumps using CFX

HONORS & AWARDS

South China University of Technology Graduate Scholarship (three times, RMB18,000)

Sep. 2022 – Jun. 2025

Merit Undergraduate Thesis, Northeastern University (Top 2%)

Jun. 2022

Likang Group Safety Engineering Scholarship (RMB 3,000)

Oct. 2021

The National Undergraduate Mathematics Competitions, Third prize of national

Dec. 2020

The National Undergraduate Mathematics Competitions, Second prize of provincial

Dec. 2020

Northeastern University Merit Student Scholarship, Third prize (four times)

Sep. 2018 – Jun. 2022

ACADEMIC CONFERENCES

The 4th International Conference on Fluid and Chemical Engineering, Wuhan, China

July 26 – 27 2024

The 3th International Conference on Fluid and Chemical Engineering, Wuhan, China

July 21 – 22 2023

SKILLS

Technology: ANSYS, COMSOL Multiphysics, OLGA, MATLAB, SpaceClaim

Language: Mandarin (Native), English (IELTS: 6.5), Cantonese (Basic)